

PROJECT REPORT: HOSPITAL AGENTIC AI

TASK 1: Core Implementation & Architecture

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1 PROJECT OVERVIEW

This project implements a customer-facing **Agentic AI Assistant** capable of managing hospital appointments. Task-1 establishes the foundational layer using a **Multi-Agent Architecture** backed by **SQLite** and **Rule-Based NLP**.

- **Objective:** Automate doctor availability checks and appointment bookings.
- **Tech Stack:** Python, SQLite, Regex-based NLP.
- **Key Feature:** Atomic Booking System with Conflict Resolution.

2 SYSTEM ARCHITECTURE (MULTI-AGENT)

The system is modularized into specialized agents. A high-level data flow diagram is shown below.

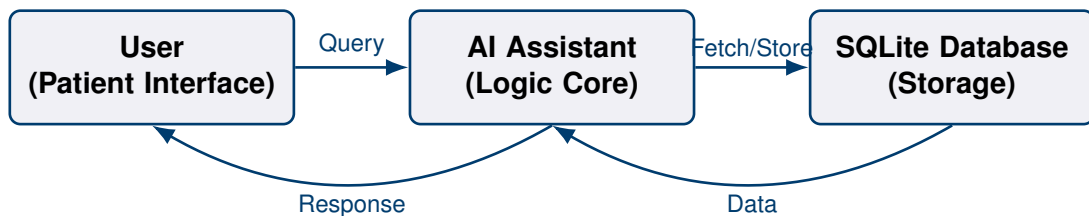


Figure 1: High-Level Data Flow Architecture

Agent Responsibilities:

Authentication Agent	Securely handles login (Test User: <code>patient1</code>).
Patient Profile Agent	Captures runtime patient data (Name, Age, Issues) into DB.
Intent Classifier	(<i>Rule-Based</i>) Routes queries: Symptoms → Availability → Booking.
DB Query Agent	Executes optimized SQL queries to fetch slots and commit bookings.
Response Agent	Formats raw data into human-readable, polite responses.

3 DATABASE DESIGN (SQLITE)

Table 1: doctors_availability

Column	Description
doctor_id	Primary Key (Auto)
department	Cardiology, Neuro, etc.
time_slot	e.g., 09:00-10:00
is_booked	Boolean Status

Table 2: patients

Column	Description
patient_id	Primary Key
name	Patient Full Name
medical_issue	Symptoms stored
age	Integer value

4 CORE LOGIC & NLP STRATEGY

Why Rule-Based for Task 1?

We utilized keyword mapping and Regex for intent classification to ensure **determinism** and **zero-latency** for the initial prototype.

- **Input:** "I have chest pain" → **Logic:** Keyword "chest" → **Dept:** Cardiology.
- **Input:** "Book Dr Rahul" → **Logic:** Extract Entity "Rahul" → SQL Transaction.

5 SYSTEM EXECUTION OUTPUT

Below is a real interaction log captured from the terminal session.

Terminal Output Log

```
> Username:  patient1
> Password:  *****
[SUCCESS] Login Verified.

> Enter Name:  Kaushal Raj | Age:  22 | Issue:  chest pain
[INFO] Profile Saved.

> User:  Show available doctors in Cardiology on Monday
> Agent:  Dr Rahul is available at 09:00-10:00.

> User:  Book a slot with Dr Rahul
> Agent:  [CONFIRMED] Appointment ID #102 created for Monday.
```

6 CONCLUSION & FUTURE SCOPE

Task-1 successfully delivers a **robust backend** with persistent storage. The simple architecture allows seamless swapping of the Rule-Based Agent with an **LLM/BERT model** in Task-2 without altering the database layer.